

A Field reference (80 sites)

Campaign

82 site visits, five neighbourhoods (LCZ 1–5)
20, 23, 25, 26 August 2026, 11:19–15:06
solar elevation 46–67°; T_a 32.0–38.7 °C
80 sites enter the analysis (one panorama, one
globe reading missing)

Instruments at 1.5 m, same minute

globe thermometer Ø 0.05 m, ϵ 0.95 → T_g
weather meter → T_a , RH, wind
360° camera → two fisheye images
thermal camera → 328 radiometric images
(324 decoded), wall and pavement

Panorama processing

stitch with per-image attitude correction;
project to 4 horizontal + 4 tilted + nadir faces
SegFormer-B0 (ADE20K) → sky, tree, building,
residual (median 0.02); 1° zenith–azimuth grid,
Steyn (1980) sin 2θ hemispheric integration
→ SVF, TVF, BVF, sum = 1
one pose failure → 79 sites

Sun or shade at the globe

two independent readers on the same
panorama, 76 of 80 agreed; disagreements
settled by the globe rise; cloud-diffuse = shade
→ 62 sunlit, 18 shaded
check: globe rise 21.7 K sunlit vs 9.4 K shaded

Reference T_{mrt} and PET

ISO 7726 forced convection:
 $T_{mrt} = [(T_g + 273)^4 + 1.1 \cdot 10^8 \cdot \gamma^{0.6} (\epsilon D^{0.4}) (T_g - T_a)]^{1/4} - 273$
PET: pet_steady, 1.4 met, 0.9 clo, sitting
→ 35.9–53.8 °C; 63 of 80 at ≥ 41 °C

Field reference

view factors (79) · sun or shade (62/18)
surface temperature · reference PET (80)
the reference for every comparison;
not an input to the service

on-site sun or shade, T_a , wind;
reference PET = training target

B Imagery-free service chain — as deployed

Building register

footprint polygons + floors
 $H = 3.018 \cdot \text{floors} + 0.902$ m
(no floors → 2 storeys)

Canopy height raster

Meta/WRI 1 m → 30 m cells
effective screening $(1 - \tau)$,
 $\tau = 0.35$

Gridded weather

T_a , RH, wind speed and
direction at the site's cell;
cloud fraction

Geometry at the pedestrian (eye height 1.5 m)

snap the point outside buildings and to the street centre line (≤ 30 m)
ray casting every 2° in azimuth to the building outlines → horizon angle $\beta(\text{az})$;
canopy above the roofline screens $(1 - \tau)$; SVF = $1 - \text{mean sin}^2\beta$ (Steyn 1980); BVF
street width and H/W from the narrowest opposing pair (5° rays, ≤ 60 m)
sun blocked? horizon[sun azimuth] > sun elevation → direct-beam shade 0/1
(buildings only; tree shade off in Korea)
dominant wall material → wall albedo and emissivity (height / distance weighted)

SVF verified against the panorama SVF at the 79 field sites: MAE 0.112, bias +0.046, r 0.70

Solar position and irradiance

pvlb; NREL SPA position;
Ineichen–Perez clear sky
→ Kasten & Czeplak (1980)
cloud attenuation, a 0.75, b 3.4
→ Erbs et al. (1982) diffuse
fraction → GHI, DNI, DHI
beam turbidity 0.6 (Busan haze)

Pedestrian wind at 1.5 m (PWI)

reference wind → von Mises weighting of wind direction against the street axis (κ 2)
× exponential partial coefficients for sky opening (SVF), building and vegetation
screening → u_p (bypassed when an on-site wind reading is supplied)

Ground surface energy balance → T_s

$(1 - \alpha) \cdot S_i \cdot (1 - f_{\text{stor}}) + \epsilon_g \cdot (L_i - \sigma T_s^4) - h_c \cdot (T_s - T_a) = 0$, solved by Newton iteration
 $S_i = \text{DNI} \cdot \sin\beta \cdot \text{shade} + \text{DHI} \cdot \text{SVF}$, with a 2 h exponential lag and a 40 W m⁻² storage release; $L_i = \sigma T_{sky}^4 \cdot [\text{SVF} \cdot \epsilon_{sky} + (1 - \text{SVF}) \cdot \epsilon_{\text{env}}]$
 $h_c = a + b \cdot u_p$; α , ϵ_g from Sentinel-2 surface class (default 0.15 / 0.95); shaded ground keeps 0.74 of its sunlit rise; vegetated share → T_g

Six-direction radiation budget → T_{mrt} (VDI 3787 Part 2)

$S_{dir} = a_L \cdot [f_p(\beta) \cdot \text{DNI} \cdot \text{shade} + \Sigma_i F_i \cdot (\text{DHI} \cdot \psi_{sky,i} + \alpha \cdot \text{GHI} \cdot \psi_{grd,i})] + \epsilon_p \cdot \Sigma_i F_i \cdot (L_{sky,i} \cdot \psi_{sky,i} + L_{surf} \cdot \psi_{grd,i})$
 a_L 0.7, ϵ_p 0.97; $F = 0.22 \times 4 \text{ sides} + 0.06 \text{ up} + 0.06 \text{ down}$; $f_p(\beta) = 0.308 \cdot \cos[\beta(0.998 - \beta/50000)]$ (Fanger 1970)
 $\psi_{sky} = \text{SVF up, SVF/2 sideways}$; $L_{sky} = \epsilon_{sky} \sigma T_{sky}^4$, ϵ_{sky} by Brunt (1932) + Crawford & Duchon (1999); $L_{surf} = \epsilon_g \sigma T_s^4$
sunlit walls reflect short-wave (wall albedo × 0.45 sunlit fraction); $T_{mrt} = (S_{dir} / \epsilon_p \sigma)^{1/4} - 273.15$

PET, physics — MAE 6.6 °C against the field reference (gridded inputs, all 80 sites)

pythermalcomfort pet_steady (Höppe MEMI, VDI 3787 Part 2): T_a , T_{mrt} , u_p , RH; 1.37 met, 0.5 clo, standing
29 of 63 extreme-heat sites detected

Residual correction layer — the only learned step

five predictors, standardised: sun or shade, geometric SVF, T_a , wind, PET_{phys} — no coordinates
ridge regression, $\alpha = 20$, trained on the 80 field sites against the reference PET; Δ = residual estimate
confidence c from the standardised distance to the training set: $c = 1$ within RMS-z 1.5, falling to 0 at 4 (physics fallback)
PET = PET_{phys} + $c \cdot \Delta$; evaluated leave-one-neighbourhood-out (the whole neighbourhood held out)

Reported PET

MAE 1.8 °C, bias −0.0 °C, r 0.82 against the field reference (leave-one-neighbourhood-out)
all 63 extreme-heat sites (PET ≥ 41 °C) detected



measured or reported data



computation



learned step



external dataset

field data in

--- compared